



Shares handled by Blockchain

Maintaining share records has gotten a lot easier and safer thanks to StockChain, a recently launched shares registry. <https://digit.in/oct18-22-1>



Amazon aim for cashier-less stores

Amazon plans to open 3000 cashier-less Amazon Go convenience stores by 2021. <https://digit.in/oct18-22-2>

A story in sound

From cylindrical tubes to portable music players, sound recording and playback has a long story to tell

Arnab Mukherjee | arnab@digit.in

S

ince the invention of language, there's been a consistent need to record human expression. Why

wouldn't there be? Once we had mastered communication, it is only natural that the human mind desired to preserve its thoughts, memories and experiences in ways that could, at the very least, outlast human memory. The first step in this direction was the written word, and that was pretty much it for almost the entirety of known human history. It is only in the 1800s that we started making progress towards preserving sound. Today, digital files smaller than a photograph can play an orchestra right into your ears. How did we get from scratching lines onto a tube to this?

PURE ACOUSTICS

To peg the beginning of our efforts to record sound, we have to go as far back as the 1850s. The phonautograph was patented in 1857 by Parisian inventor Édouard-Léon Scott de Martinville. The recorded form was never intended for playback – a stylus would mark sound-wave modulated lines on a coating of soot on white paper as the paper passed underneath, based on vibrations generated by a sound wave.



Thomas Edison and his early phonograph

The first practical device for recording and playback of sound was invented by the man of a thousand patents, Thomas Edison. His invention, the phonograph, consisted of a cylinder with a tinfoil wrapped around it, and a stylus connected to a diaphragm. The cylinder would move laterally when a handle was turned, etching onto the tinfoil the approximate pattern of vibrations created by the pressure of the sound waves hitting the diaphragm. Playback was just the opposite - rotating the cylinder the opposite way would vibrate the diaphragm generating the sound again. In theory, this was a highly practical idea, but practically, the recordings were barely audible. Improvements over the next decade incorporated a hard wax cylinder for greater accuracy. However, the next major jump in this technology came to be as a way of avoiding conflict with Edison's patent.

THE GRAMOPHONE AND THE RECORD

Between 1887 and 1893, Emile Berliner came up with the flat record and the gramophone. Contrary to Edison's

cylinder, Berliner's recording medium was a flat disc coated with lampblack. A thin stylus moved laterally (instead of vertically as in Edison's design) etching a pattern based on the pressure created by a vibrating diaphragm. After the etching was done, the disc was given an acid bath to etch the grooves into it and was ready for playback. Berliner soon realised that instead of etching, using an electroplating technique to create ridges instead of groove would create a stamper that could be used for mass production in a steam heated process. Replacing the acid etching system with

wax also improved fidelity and pushed records and gramophones to the masses.

As compared to the cylinders, discs were more than just a patent alternative. Marginally louder than cylinders, discs



A gramophone

also allowed recording on both sides of the surface. Consequently, sales of discs had overtaken cylinders by 1910 and by the time the First World War was over, discs had completely

taken over the market. Various formats and materials were tried out to improve the durability and bring down the cost. Eventually, vinyl records with a diamond stylus became the standard.

ENTER ELECTRICITY

A new era of sound recording was ushered in when electricity entered the fray. Even though the main recording mechanism was still mechanical for